

Lab 01: Transforming Resume Analyzer to a Web Application

To transition a Python-based CLI Resume Analyzer into a functional web application using **Streamlit** and deploy the final product to **Streamlit Cloud**.

Task 1: Mapping CLI Logic to UI Components

In Day 4, your application (Resume Analyzer) relied on `input()` and `print()` statements. In a web context, these must be replaced with interactive widgets. Modify your script to implement the following:

- Create an `app.py` file to start your work (all the UI components are here).
- **File Upload:** Replace file path strings with `st.file_uploader` to accept PDF.
- **Job Description Input:** Use `st.text_area` for students to paste the target job requirements.
- **Analysis Trigger:** Wrap your processing logic inside a `st.button("Analyze Resume")`.
- **Results Display:** Use `st.success()`, `st.write()`, or `st.table()` to display the match score and extracted keywords.
- Modification is needed on certain processes (as it may not be required when you convert the CLI application to a web application).
- You may use local AI model or any free online AI model.
- For hosting part, there will be no AI feature if a local AI model is used in the project and that is totally fine in this lab, as long as your application is hosted and can be accessed online.

Task 2: Preparing for Deployment

To host your app on Streamlit Cloud, your project folder must contain:

1. `app.py`: The main entry point of your Streamlit application.
2. `requirements.txt`: A list of all dependencies

Task 3: Deployment to Streamlit Cloud

1. Refer to the notes: **05_Cloud_Hosting**.

Submission Instructions

Once you have completed your code refactoring, complete the following two deployment steps:

1. **GitHub:** Upload all your refactored project files and folders to a GitHub repository (ensure your `.env` and `.db` files are ignored!).
2. **Streamlit Cloud:** Host your application live on Streamlit Cloud, making sure to add your `GEMINI_API_KEY` to the Advanced Secrets panel.

What to Submit:

Create and submit a single plain text file named `lab01_yourname.txt` using the exact layout below. Do **not** upload your raw python files directly to the learning management system.

Plaintext

```
1  =====
2  STUDENT NAME      : [Enter Your Full Name Here]
3  LAB ASSIGNMENT    : Lab 01
4  =====
5
6  [GITHUB REPOSITORY LINK]
7  https://github.com/your-username/your-repository-name
8
9  [STREAMLIT LIVE APPLICATION LINK]
10 https://your-custom-subdomain.streamlit.app/
11
12 =====
```